

Find and Fix: Inconclusive Tests and Forgotten Endpoints

Directions

- Every problem shows student work that is wrong in one specific way. Repair the computation, then name the rule that was broken.
- Two errors dominate this topic: treating an *inconclusive* test as if it had reached a conclusion, and stopping at the open interval a ratio test produces without testing the endpoints.
- Part (a) always wants a value; put it on the Answer line. The remaining parts want a sentence or two, written in the work space.

1. A student writes: “For $\sum_{n=1}^{\infty} \frac{1}{n}$, the terms satisfy $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, so by the nth-term test the series converges.”

- (a) Confirm the limit the student computed.
- (b) Name the error, and state what the nth-term test does and does not prove.

Answer: _____

2. For $\sum_{n=1}^{\infty} \frac{x^n}{n 2^n}$ a student computes the radius correctly and then writes: “so the interval of convergence is $(-2, 2)$.”

- (a) Confirm the radius of convergence, using the coefficient ratio $\frac{2(n+1)}{n}$.
- (b) Write the exact sum of the series at the left endpoint $x = -2$.
- (c) Name the error and give the correct interval of convergence.

Answer: _____

3. For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ a student estimates the sum with the first 8 terms and reports the error bound as $\frac{1}{64} \approx 0.0156$.

- (a) Give the correct error bound as an exact fraction.
- (b) Name the error the student made.

Answer: _____

4. A student computes $\sum_{n=2}^{\infty} \left(\frac{1}{4}\right)^n$ as $\frac{1}{1 - \frac{1}{4}} = \frac{4}{3} \approx 1.333$.

- (a) Give the correct sum as an exact fraction.
- (b) Name the error the student made.

Answer: _____

5. A student writes: “For $\sum_{n=1}^{\infty} \frac{n^2}{n^3 + 1}$ the ratio test gives a limit of 1, so the series converges.”

- (a) Confirm the ratio-test limit.

Answer: _____

- (b) Find $\lim_{n \rightarrow \infty} \frac{n^2/(n^3 + 1)}{1/n}$, the limit comparison against the harmonic series.

Answer: _____

- (c) Name the error and give the correct conclusion.

6. For $\sum_{n=1}^{\infty} \frac{(2x-6)^n}{n}$ a student reports a radius of 1 and an interval of $(2, 4)$.
- (a) Rewrite $(2x-6)^n$ as $2^n(x-3)^n$, so the coefficients are $c_n = 2^n/n$ with ratio $\frac{n+1}{2n}$, and give the correct radius.
- (b) Name both faults in the student's answer and give the correct interval of convergence.

Answer: _____

7. A student writes: " $\int_1^{\infty} x^{-3/2} dx = [-\frac{2}{5}x^{-5/2}]_1^{\infty} = \frac{2}{5}$, so by the integral test $\sum_{n=1}^{\infty} n^{-3/2}$ converges and its sum is $\frac{2}{5} = 0.4$."
- (a) Evaluate the integral correctly.
- (b) Name *both* errors in the student's sentence.

Answer: _____

8. Here is a student's complete answer for $\sum_{n=1}^{\infty} \frac{(x+2)^n}{n 3^n}$: "The ratio test gives $\frac{|x+2|}{3} < 1$, so $-5 < x < 1$ and the interval of convergence is $(-5, 1)$."

(a) Substitute the left endpoint $x = -5$ and write the exact sum of the resulting series.

Answer: _____

(b) Substitute the right endpoint $x = 1$ and give $\lim_{n \rightarrow \infty}$ of the resulting terms.

Answer: _____

(c) Name the error and give the correct interval of convergence.